

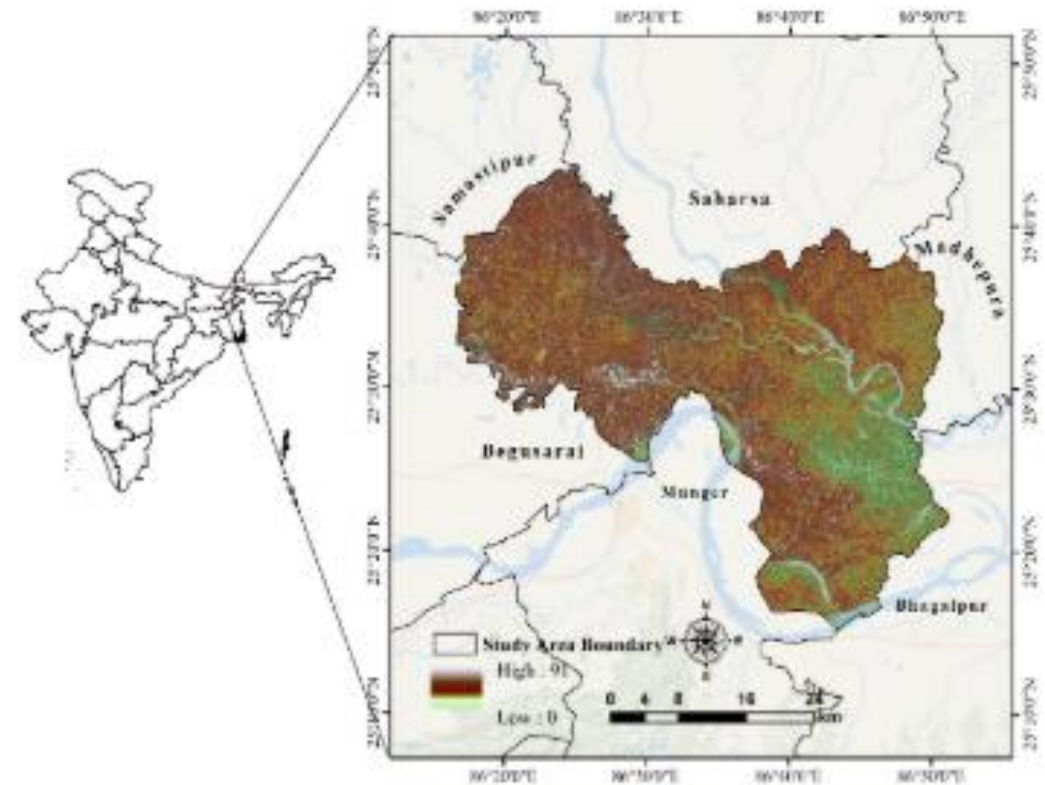
Spatio-temporal Analysis of Land Use and Land Cover Changes in Khagaria District, Bihar (2015–2025) using Remote Sensing and GIS Techniques and Future prediction of 2035

Land Use and Land Cover (LULC) dynamics play a crucial role in understanding the interaction between human activities and natural processes, particularly in flood-prone regions. Khagaria District is highly vulnerable to recurrent flooding, which significantly alters its landscape and agricultural patterns. This study applies Remote Sensing and GIS techniques to analyze spatio-temporal LULC changes from 2015 to 2025. It further integrates predictive modelling using the CA-Markov approach to simulate future land-use scenarios for 2035. The findings aim to support sustainable land management and informed decision-making in flood-affected regions.

Aims, Objective, Study Area and Dataset

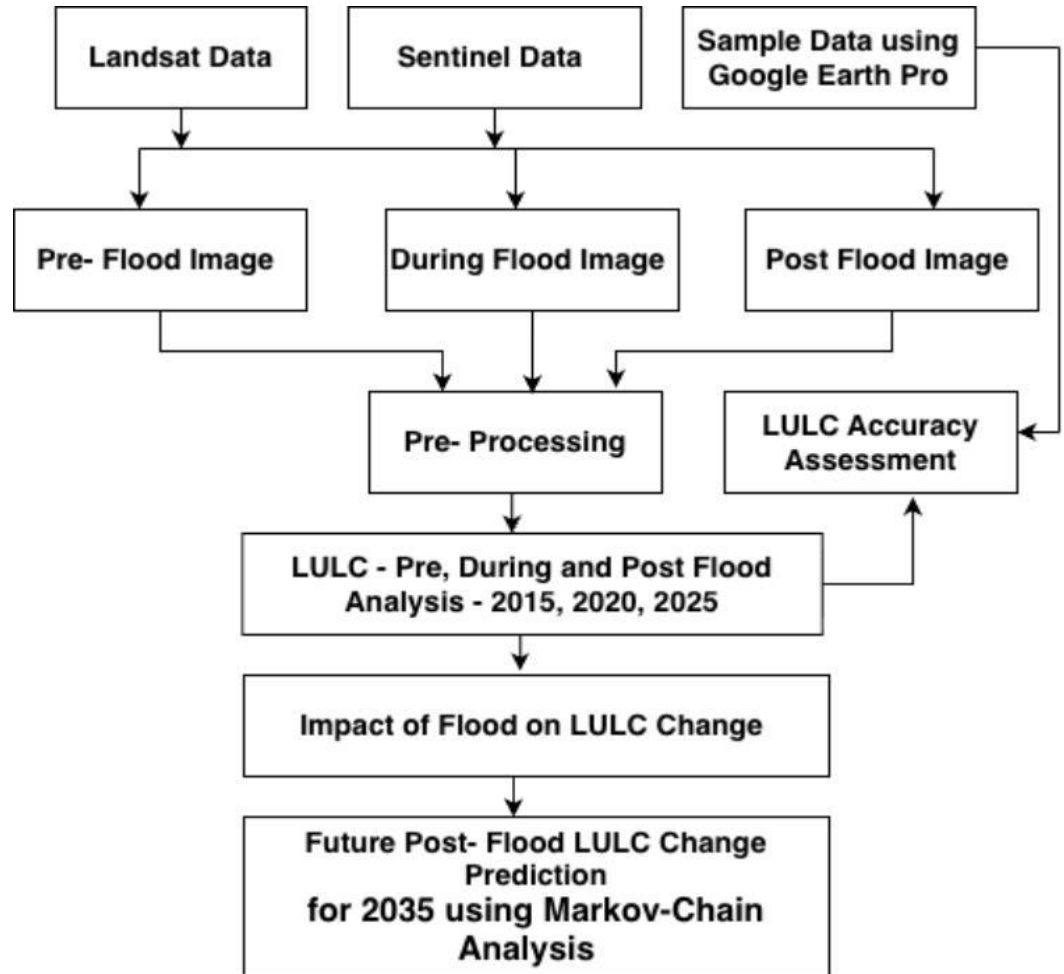
Aim and objective-To assess future post-flood LULC changes and simulating land-use changes under different scenarios in the Khagaria district using the cellular automata (CA)–Markov model.

SNo.	Data Type	Source
	Landsat 8 (Operational Land Imager & TIRS) (30 m spatial resolution) (2015)	USGS Earth Explorer
2.	Sentinel - 2 (10 m spatial resolution) (2020,2025)	Copernicus Open Access Hub
3.	Sentinel - 1 (10 m spatial resolution)	Copernicus Open Access Hub
4.	OSM Toposheet No.- 72K/11, 72K/10, 72K/6, 72K/7, 72K/14, 72K/15 (2010)	Survey of India (SOI)
5.	Google Earth	https://earth.google.com/web/



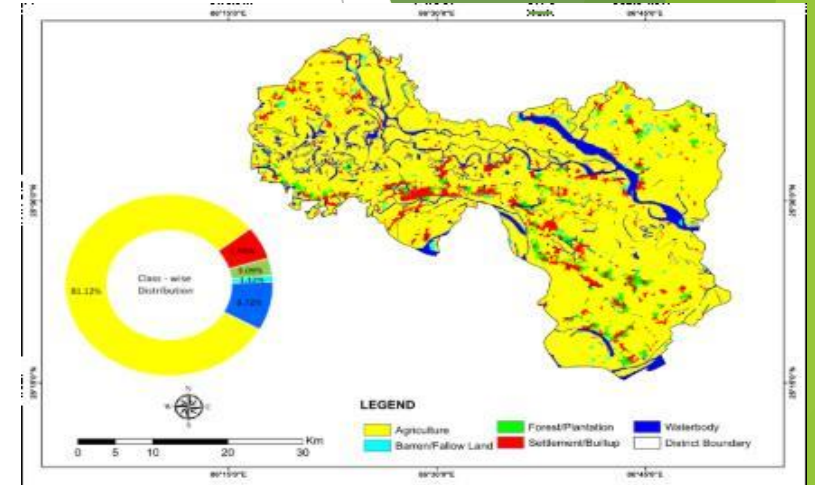
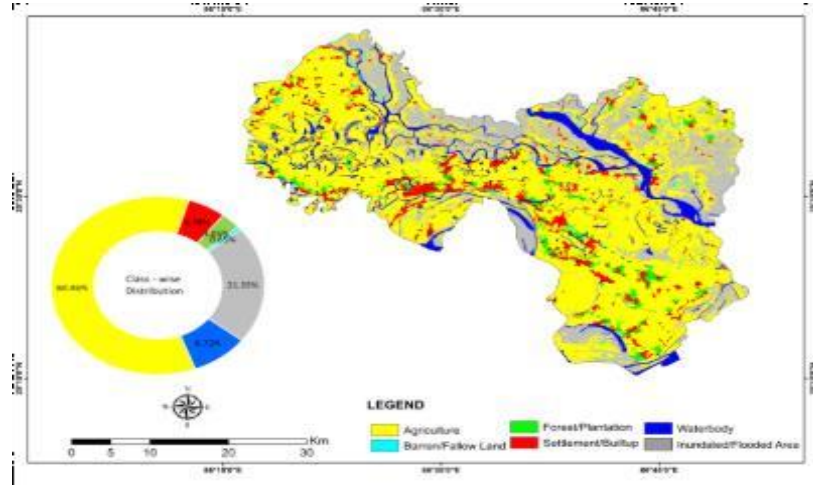
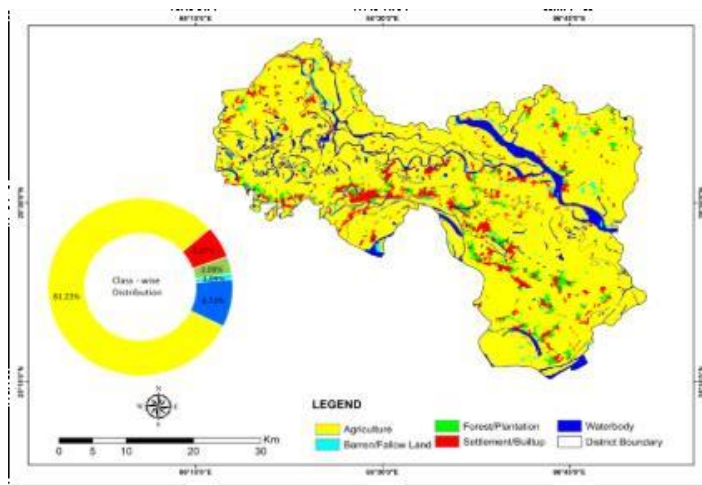
Study Area Map : Khagaria

Methodology



- ▶ Satellite data from Landsat, Sentinel, and Google Earth are used to generate pre-, during-, and post-flood images.
- ▶ These images undergo preprocessing and accuracy assessment to ensure reliable classification. LULC analysis is then conducted for 2015, 2020, and 2025 to evaluate flood impacts on land cover changes. Finally, future LULC changes for 2035 are predicted using the Markov Chain model.

LULC Maps : 2015



Yellow: Cropland
Red: Built-up area
Blue: Water
Green: Vegetation
Aqua: Bareground
Grey: Inundation

Figure 3. LULC Khagaria district, 2015, before flood, Figure 4. LU LC Khagaria district, 2015, during flood. Figure 5. LULC Khagaria district, 2015, after flood

LULC Maps : 2020

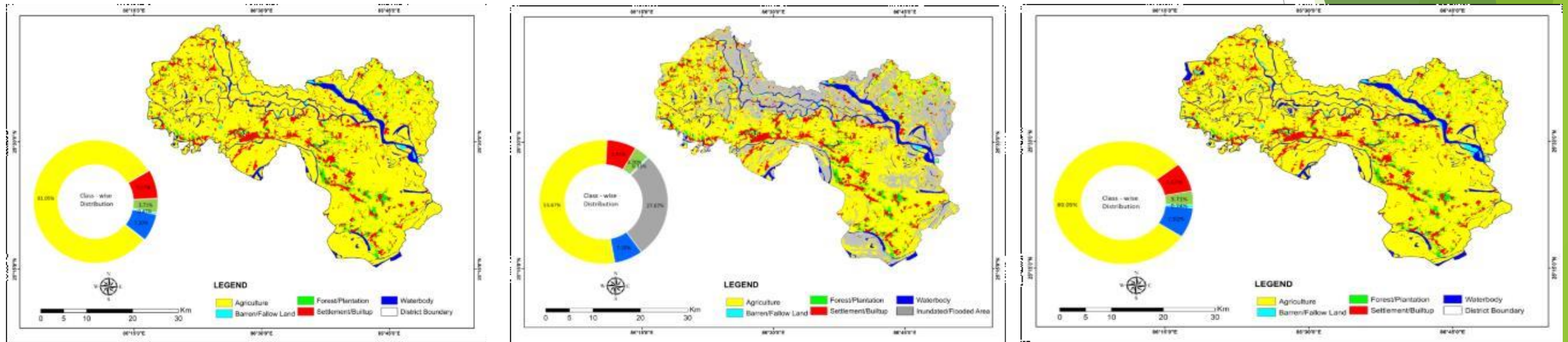


Figure 1. LULC Khagaria district, 2020, before flood, Figure 2. LULC Khagaria district, 2020, during flood. Figure 3. LULC Khagaria district, 2020, after flood

LULC Maps : 2025

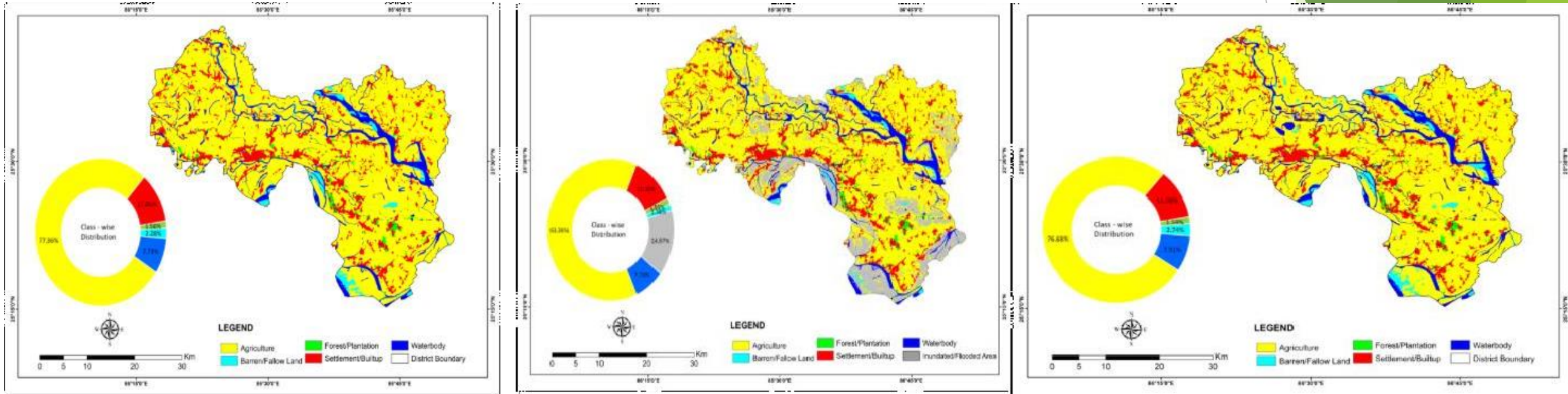
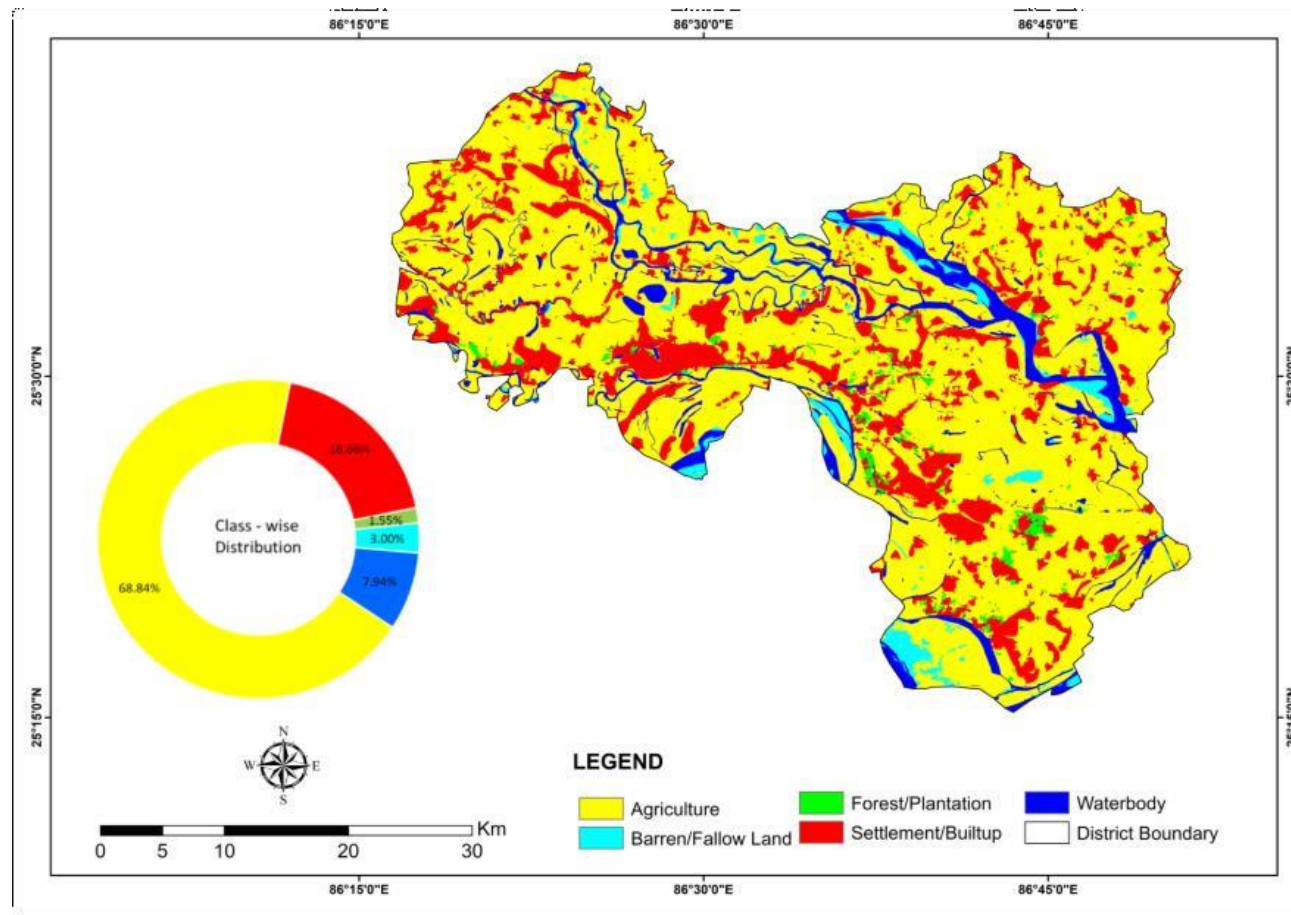


Figure 1. LULC Khagaria district, 2025, before flood, Figure 2. LU LC Khagaria district, 2025, during flood. Figure 3. LULC Khagaria district, 2025, after flood

LULC Maps : 2035 (Predicted)



Future LULC Khagaria district, 2035 created by CA-Markov Chain Modelling

Conclusion

- ▶ Key Findings
 - ▶ LU/LC change analysis conducted for 2005-2025 using Landsat 8 and Sentinel-2 data
 - ▶ Five classes identified: built-up, farmland, barren land, water bodies, wetlands
 - ▶ High classification accuracy: 95.24% (2015) and 95.0% (2025)
 - ▶ Agricultural land decreased by 3.87% due to flood impacts
 - ▶ Built-up area increased by 5.19%, including expansion into flood-prone zones
 - ▶ Water bodies decreased by 0.99%
 - ▶ Barren/fallow land increased by 1.19%
 - ▶ Forest/plantation area declined by 1.53%
 - ▶ Flood identified as the major driver of LU/LC changes
- ▶ Why It Matters
 - ▶ Highlights the significant impact of floods on land transformation in Khagaria
 - ▶ Shows risk to agriculture, the dominant livelihood sector
 - ▶ Indicates unsafe expansion of settlements into flood-prone areas
 - ▶ Reflects environmental degradation (loss of forest and water bodies)
 - ▶ Provides scientific evidence for planning and disaster management
 - ▶

Recommendations

- ▶ Implement **risk-based land-use planning** to restrict development in flood-prone zones
- ▶ **andatory insurance policies** for high-risk assets
- ▶ Promote **sustainable agricultural practices** to reduce flood vulnerability
- ▶ Strengthen **flood management and mitigation strategies**
- ▶ Use LU/LC analysis for **identifying drivers of land degradation and planning interventions**